

PHY 303- Electrodynamics Mid Semester Examination II

Please write short and concise answers. Illegible answers will not be graded.
Maximum marks = 25, Time = 60 minutes

1. Expand the following function in terms of Legendre Polynomials and write down the first two terms [5]

$$f(x) = \begin{cases} +1 & \text{for } x > 0 \\ -1 & \text{for } x < 0 \end{cases}$$

2. The lowest order term in the vector potential due to a current carrying loop in the xy plane is given by

$$A_\varphi = \frac{\mu_0 I a^2 r \sin \theta}{4(a^2 + r^2)^{3/2}}$$

Find the components of the magnetic field. [5]

3. A long solid dielectric cylinder of radius a is permanently polarized so that the polarization is radially outward everywhere, with a magnitude proportional to the distance from the axis of the cylinder, i.e.,

$$\vec{P} = \frac{1}{2} P_0 r \hat{r}$$

Find the volume charge density in the cylinder. [5]

4. A charge q is placed at a distance d from a dielectric sphere of radius a with permittivity ϵ . Assume that the line joining the centre of the sphere to the charge is along the z-axis. An image charge iq is at $z = a^2/d$.

- Write the expression for the electrostatic potential outside the sphere and expand the potential for the regions $r < d$ and $r > d$ in terms of Legendre polynomials.
- Put an image charge q'' at $z = d$ and write down the expression for the potential inside the sphere, and expand in terms of Legendre polynomials (Hint: Remember to use the correct permittivity in all the cases).
- Using the boundary conditions

$$(\vec{D}_2 - \vec{D}_1) \cdot \hat{n} = \sigma, \quad (\vec{E}_2 - \vec{E}_1) \times \hat{n} = 0,$$

find the charges q'' and q' in terms of q (Hint: Write down the boundary conditions in terms of derivatives of Φ). [3+3+4]

Useful Formulae:

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn}$$

Divergence in cylindrical and spherical coordinates:

$$\frac{1}{\rho} \frac{\partial (\rho A_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z}$$

$$\frac{1}{r^2} \frac{\partial (r^2 A_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\varphi}{\partial \varphi}$$

Curl in cylindrical and spherical coordinates:

$$\left(\frac{1}{\rho} \frac{\partial A_z}{\partial \varphi} - \frac{\partial A_\varphi}{\partial z} \right) \hat{\rho} + \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) \hat{\varphi} + \frac{1}{\rho} \left(\frac{\partial (\rho A_\varphi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \varphi} \right) \hat{z}$$

$$\frac{1}{r \sin \theta} \left(\frac{\partial}{\partial \theta} (A_\varphi \sin \theta) - \frac{\partial A_\theta}{\partial \varphi} \right) \hat{r} + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \varphi} - \frac{\partial}{\partial r} (r A_\varphi) \right) \hat{\theta} + \frac{1}{r} \left(\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right) \hat{\varphi}$$